Transmission, Narrative Variation, and the Limits of Reconstruction: An Isnād-cum-Matn Study of the Ḥadīth of al-Jassāsah

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ABSTRACT

*The ḥadīth of al-Jassāsah recounts an island encounter with an enigmatic informant and a chained figure identified as the Dajjāl, embedded in reports of a prophetic sermon narrated by Fāṭimah bint Qays. This study was carried out as an empirical proof-of-concept to evaluate the combined functionality of **Masar** as an automated takhrīj and pericope-alignment engine and **Riwaq** as an Isnād-cum-Matn analysis (ICMA) and stemma engine. Analyzing a primary corpus of twenty records yielding twenty-two distinct compiler paths across nine classical collections, the pipeline establishes ʿĀmir al-Shaʿbī (d. ca. 103–105 AH / 721–723 CE) as the primary structural common link (Conf = 0.6743, bundle coverage 86.4%, bypass ratio 4.5%), with collateral branches through Abū Salamah (Medina) and Yaḥyā ibn Yaʿmar (Khurasan/Basra) demonstrating early multi-regional transmission beyond Kufa. Textual collation across seven narrative episodes reveals three traveler profiles—Tamīm sailing personally with Lakhmid/Judhamite companions, Tamīm's kinsmen sailing without him, and unnamed regional voyagers—alongside three distinct lexical registers for al-Jassāsah: a zoomorphic beast (dābbah / al-Jassās), an ambiguous hair-covered human (insān / shayʾ), and an explicitly female personification (imraʾah / jāriyah), the latter concentrated in collateral lines. Crucially, automated pericope alignment demonstrates that the kinsmen reading (banū ʿamm) is attested across four distinct pupil branches radiating from al-Shaʿbī, refuting the classical suspicion that this formulation was an idiosyncratic corruption exclusive to Mujālid ibn Saʿīd. Within the ICMA framework, this establishes a conditional transmission horizon in late first-century Kufa, distinct from third-century literary compilation dates and later physical manuscript survival, demonstrating the viability of a coupled takhrīj-ICMA computational architecture for scalable hadith studies.*

Introduction

In its most familiar form, the Jassāsah narrative is embedded within Fāṭimah bint Qays's recollection of a public sermon. The Prophet relates news brought by Tamīm al-Dārī, whose maritime story features a strange hairy being, a chained man, questions about places in Palestine and its environs, and the latter's identification as the Dajjāl. The long version in *Ṣaḥīḥ Muslim* connects the account with questions about Mecca and Medina and a gesture pointing east [1]. Yet the surviving reports differ in their framing, participants, scene descriptions, sequence of interrogations, and endings. Reading one long account as the fixed original would conceal those differences.

The historical question is consequently twofold: what transmission structures do the extant witnesses actually report, and which textual features correlate with those structures? A common-link analysis can describe branching among named transmitters; an isnād-cum-matn analysis must additionally test whether particular readings travel with those branches. Neither operation by itself establishes the reported event as an independently observed occurrence or determines the date at which a detail first circulated.

Corpus and method

This investigation synthesizes a primary source-text corpus and source audit comprising nineteen separately indexed digital attestations: Muslim 2942a–d; Abū Dāwūd 4325–4328; al-Tirmidhī 2253; Ibn Mājah 4074; al-Nasāʾī's *al-Sunan al-Kubrā* 4453 in the consulted display; Aḥmad 26066 in the Wikisource numbering; Ibn Abī Shaybah 40425; Ibn Ḥibbān 6787; al-Ṭayālisī 1751; and al-Ṭabarānī's *al-Muʿjam al-Kabīr* 956, 960, 967 and 968 [1–12]. The primary source audit preserves source-form Arabic, normalized isnāds, and verified digital locators across all nineteen attestations. Collection numbering follows the displayed digital editions and may not coincide with other editions.

A *witness* or *attestation* here denotes one separately displayed entry in a collection, while a *compiler path* denotes a distinguishable isnād leading to that compiler. In the nineteen-attestation baseline, Muslim 2942a and 2942c each cite two distinct teachers within a single textual entry, yielding twenty parsable compiler paths across eighteen encoded records. Al-Ṭabarānī 967 constitutes the nineteenth textual witness; its compound isnād conjunction ("عن قتادة، عن الشعبي، وإبراهيم بن عامر، عن الشعبي") is retained for textual collation but left unparsed in the baseline graph to avoid artificial path inflation. The twenty parsed paths resolve into twelve lower-branch classes: nine pupil lines radiating from al-Shaʿbī, two collateral approaches to Fāṭimah, and a distinct Jābir comparator. Crucially, these lower branches represent downstream transmission paths from their respective authorities rather than twelve independent eyewitnesses to the Prophet.

Comparison was organized across seven canonical episodes (P_1 … P_7): the occasion of the sermon (P_1); identification of the voyagers (P_2); island landing and encounter with al-Jassāsah (P_3); description and binding of the captive (P_4); geographical and religious inquiries regarding Baysān, Lake Ṭabariyyah, Zughar, the Arabian Prophet, and in some recensions, Persia (P_5); messianic self-identification and sanctuary boundaries (P_6); and the Prophet's ratification, orientation gesture, and closing remarks (P_7). A source's cross-reference to a previously quoted longer report (e.g., *wa-dhakara al-ḥadīth*) is treated as an editorial abridgment rather than evidence that omitted episodes were unknown to its transmitter.

Source verification was conducted through digitized Arabic text editions of classical standard reference printings (specifically digitized recensions hosted on Sunnah.com, Islamweb Library, Shamela, and Wikisource, reflecting standard scholarly editions such as Fuʾād ʿAbd al-Bāqī's *Ṣaḥīḥ Muslim*, Muḥyī al-Dīn ʿAbd al-Ḥamīd's *Sunan Abī Dāwūd*, Aḥmad Shākir's *Jāmiʿ al-Tirmidhī*, Saʿd al-Shathrī's *Muṣannaf Ibn Abī Shaybah*, Shuʿayb al-Arnāʾūṭ's *Ṣaḥīḥ Ibn Ḥibbān*, and Ḥamdī ʿAbd al-Majīd al-Salafī's *al-Muʿjam al-Kabīr*). Physical paper volumes, photographic page facsimiles, and unedited manuscript folios were not directly examined in this study. A supplementary direct source verification audit incorporated two further entries from al-Ṭabarānī's *al-Muʿjam al-Kabīr*: (i) al-Ṭabarānī 964 (the Dāwūd–Ḥammād route preserving an inquest on Persia), contributing one compiler path; and (ii) al-Ṭabarānī 967, which resolves the conjoined isnād ("عن قتادة، عن الشعبي، وإبراهيم بن عامر، عن الشعبي") by encoding the primary, attested Qatādah branch as one verified compiler path (`jass-tabarani-967-route`). Rather than artificially inflating the path denominator by encoding the unverified secondary conjunct (Ibrāhīm ibn ʿĀmir) as an independent graph path, 967 is resolved to a single compiler path. Consequently, the expanded corpus consists of twenty records (18 + 2) yielding exactly **twenty-two compiler paths** (20 + 1 + 1 = 22) [14, 15].

The pipeline progression and corpus tiers are summarized in the workflow table below:

Pipeline Stage / Corpus Tier	Attestation Records	Resolved Compiler Paths	Unique Narrator Entities	Graph Edges	Analytical Treatment & Status
1. Digital Source Collation	19	—	—	—	Full transcription of 19 primary digital entries across 10 classical collections [1–12].
2. Topological Baseline Graph [13]	18	20	71	82	al-Ṭabarānī 967 compound isnād held in quarantine; unsegmented text with default matn coherence constant (0.50).
3. Masar-Enriched Baseline [15]	18	20	71	82	Active zero-based pericope spans (P_1 … P_7); empirical token-Jaccard coherence (`measured_token_jaccard` = 0.1720).
4. Expanded Model [14, 15]	20	22	77	91	Incorporates al-Ṭabarānī 964 (Dāwūd route) and resolves al-Ṭabarānī 967 (Qatādah route); empirical coherence (0.1747).

Complete Compiler Path Ledger (20 Baseline Paths → 22 Expanded Paths)

The derivation of compiler paths from primary records follows strict rules: each independent transmission route cited within a collection entry constitutes a distinct path. In the baseline corpus (18 records), *Ṣaḥīḥ Muslim* 2942a and 2942c each preserve two parallel teacher lines within a single textual record, producing four compiler paths from two entries; the remaining 16 records contain single chains (16 × 1 + 2 + 2 = 20 paths). The expanded corpus adds al-Ṭabarānī 964 (+1 path) and the resolved Qatādah branch of al-Ṭabarānī 967 (+1 path), totaling 22 compiler paths across 20 records.

#	Compiler Path ID	Primary Witness Locator	Source Collection	Authority & Intermediate Transmission Chain	Key Branch	Corpus Tier
1	`jass-muslim-2942a-awarith-route`	Muslim 2942a	*Ṣaḥīḥ Muslim*	Fāṭimah → al-Shaʿbī → Ibn Buraydah → Ḥusayn → ʿAbd al-Wārith → ʿAbd al-Ṣamad → ʿAbd al-Wārith ibn ʿAbd al-Ṣamad	Ibn Buraydah (Path A)	Baseline & Expanded

#	Compiler Path ID	Primary Witness Locator	Source Collection	Authority & Intermediate Transmission Chain	Key Branch	Corpus Tier
2	`jass-muslim-2942a-hajjaj-route`	Muslim 2942a	*Ṣaḥīḥ Muslim*	Fāṭimah → al-Shaʿbī → Ibn Buraydah → Ḥusayn → ʿAbd al-Wārith → ʿAbd al-Ṣamad → Ḥajjāj ibn al-Shāʿir	Ibn Buraydah (Path B)	Baseline & Expanded
3	`jass-muslim-2942b-sayyar-route`	Muslim 2942b	*Ṣaḥīḥ Muslim*	Fāṭimah → al-Shaʿbī → Sayyār Abū al-Ḥakam → Qurrah → Khālid → Yaḥyā ibn Ḥabīb	Sayyār Abū al-Ḥakam	Baseline & Expanded
4	`jass-muslim-2942c-hulwani-route`	Muslim 2942c	*Ṣaḥīḥ Muslim*	Fāṭimah → al-Shaʿbī → Ghaylān ibn Jarīr → Jarīr → Wahb → al-Ḥasan al-Ḥulwānī	Ghaylān ibn Jarīr (Path A)	Baseline & Expanded
5	`jass-muslim-2942c-nawfali-route`	Muslim 2942c	*Ṣaḥīḥ Muslim*	Fāṭimah → al-Shaʿbī → Ghaylān ibn Jarīr → Jarīr → Wahb → Aḥmad ibn ʿUthmān al-Nawfalī	Ghaylān ibn Jarīr (Path B)	Baseline & Expanded
6	`jass-muslim-2942d-abu-zinad-route`	Muslim 2942d	*Ṣaḥīḥ Muslim*	Fāṭimah → al-Shaʿbī → Abū al-Zinād → Mughīrah → Yaḥyā ibn Bukayr → Abū Bakr ibn Isḥāq	Abū al-Zinād	Baseline & Expanded
7	`jass-abudawud-4325-zuhri-route`	Abū Dāwūd 4325	*Sunan Abī Dāwūd*	Fāṭimah → Abū Salamah → al-Zuhrī → Ibn Abī Dhiʾb → ʿUthmān → al-Nufaylī	Abū Salamah (Medina)	Baseline & Expanded

#	Compiler Path ID	Primary Witness Locator	Source Collection	Authority & Intermediate Transmission Chain	Key Branch	Corpus Tier
8	`jass-abudawud-4326-husayn-route`	Abū Dāwūd 4326	*Sunan Abī Dāwūd*	Fāṭimah → al-Shaʿbī → Ibn Buraydah → Ḥusayn → ʿAbd al-Wārith → ʿAbd al-Ṣamad → Ḥajjāj ibn Abī Yaʿqūb	Ibn Buraydah	Baseline & Expanded
9	`jass-abudawud-4327-mujalid-sadran-route`	Abū Dāwūd 4327	*Sunan Abī Dāwūd*	Fāṭimah → al-Shaʿbī → Mujālid → Ismāʿīl ibn Abī Khālid → al-Muʿtamir → Muḥammad ibn Sudrān	Mujālid ibn Saʿīd	Baseline & Expanded
10	`jass-abudawud-4328-jabir-route`	Abū Dāwūd 4328	*Sunan Abī Dāwūd*	Jābir → Abū Salamah → al-Walīd ibn Jumayʿ → Muḥammad ibn Fuḍayl → Wāṣil ibn ʿAbd al-Aʿlā	Jābir comparator	Baseline & Expanded
11	`jass-tirmidhi-2253-qatada-route`	al-Tirmidhī 2253	*Jāmiʿ al-Tirmidhī*	Fāṭimah → al-Shaʿbī → Qatādah → Hishām al-Dastawāʾī → Muʿādh → Muḥammad ibn Bashshār	Qatādah ibn Diʿāmah	Baseline & Expanded
12	`jass-ibnmajah-4074-mujalid-route`	Ibn Mājah 4074	*Sunan Ibn Mājah*	Fāṭimah → al-Shaʿbī → Mujālid → Ismāʿīl ibn Abī Khālid → ʿAbd Allāh ibn Numayr → Muḥammad ibn ʿAbd Allāh	Mujālid ibn Saʿīd	Baseline & Expanded
13	`jass-nasai-kubra-4453-dawud-route`	al-Nasāʾī 4453	*al-Sunan al-Kubrā*	Fāṭimah → al-Shaʿbī → Dāwūd ibn Abī Hind → Ḥammād ibn Salamah → Ḥajjāj ibn Muḥammad → Muḥammad ibn al-Muthannā	Dāwūd ibn Abī Hind	Baseline & Expanded

#	Compiler Path ID	Primary Witness Locator	Source Collection	Authority & Intermediate Transmission Chain	Key Branch	Corpus Tier
14	`jass-ahmad-27102-dawud-route`	Aḥmad 26066	*Musnad Aḥmad*	Fāṭimah → al-Shaʿbī → Dāwūd ibn Abī Hind → Ḥammād ibn Salamah → ʿAffān ibn Muslim	Dāwūd ibn Abī Hind	Baseline & Expanded
15	`jass-ibn-abi-shaybah-40425-mujalid-route`	Ibn Abī Shaybah 40425	*Muṣannaf Ibn Abī Shaybah*	Fāṭimah → al-Shaʿbī → Mujālid → Ḥammād ibn Usāmah (Abū Usāmah)	Mujālid ibn Saʿīd	Baseline & Expanded
16	`jass-ibn-hibban-6787-yahya-route`	Ibn Ḥibbān 6787	*Ṣaḥīḥ Ibn Ḥibbān*	Fāṭimah → Yaḥyā ibn Yaʿmar → Ibn Buraydah → Kahmas → ʿAwn → al-Faḍl → Hārūn ibn ʿĪsā	Yaḥyā ibn Yaʿmar	Baseline & Expanded
17	`jass-audit-y`	al-Ṭayālisī 1751	*Musnad al-Ṭayālisī*	Fāṭimah → al-Shaʿbī → Sayyār Abū al-Ḥakam → Qurrah ibn Khālid	Sayyār Abū al-Ḥakam	Baseline & Expanded
18	`jass-audit-p-956`	al-Ṭabarānī 956	*al-Muʿjam al-Kabīr*	Fāṭimah → al-Shaʿbī → Muḥammad ibn Ayyūb Abū ʿĀṣim → Abū Nuʿaym → ʿAlī ibn ʿAbd al-ʿAzīz	Abū ʿĀṣim al-Thaqafī	Baseline & Expanded
19	`jass-audit-p-960`	al-Ṭabarānī 960	*al-Muʿjam al-Kabīr*	Fāṭimah → al-Shaʿbī → Abū Isḥāq al-Shaybānī → Muḥammad ibn Fuḍayl → ʿUthmān ibn Abī Shaybah → al-Ḥusayn ibn Isḥāq	al-Shaybānī	Baseline & Expanded

#	Compiler Path ID	Primary Witness Locator	Source Collection	Authority & Intermediate Transmission Chain	Key Branch	Corpus Tier
20	`jass-audit-p-968`	al-Ṭabarānī 968	*al-Muʿjam al-Kabīr*	Fāṭimah → al-Shaʿbī → Sayyār Abū al-Ḥakam → Qurrah → Ḥammād ibn Masʿadah → Ḥabīb ibn Bishr → Aslam al-Wāsiṭī	Sayyār Abū al-Ḥakam	Baseline & Expanded
21	`jass-tabarani-964-route`	al-Ṭabarānī 964	*al-Muʿjam al-Kabīr*	Fāṭimah → al-Shaʿbī → Dāwūd ibn Abī Hind → Ḥammād ibn Salamah → Musaddad → Muʿādh ibn al-Muthannā	Dāwūd ibn Abī Hind	Expanded Model (+1)
22	`jass-tabarani-967-route`	al-Ṭabarānī 967	*al-Muʿjam al-Kabīr*	Fāṭimah → al-Shaʿbī → Qatādah → Saʿīd ibn Bashīr → al-Walīd → Ayyūb → al-Ḥasan al-Maʿmarī	Qatādah ibn Diʿāmah	Expanded Model (+1)

Computational Reproducibility and Metric Formulations

To ensure methodological transparency and reproducibility, the computational pipeline integrates the Masar Takhrīj Engine with Riwaq's analytical framework under standardized schemas (`schema_version: 1`) [13, 15]:

1. **Orthographic Normalization:** Arabic source strings undergo deterministic normalization: short vowels (*tashkīl*) and tatwīl are stripped, hamzated and bare alifs are unified (`[آأإٱ] → ا`), terminal forms are harmonized (`[ىٰ] → ي`, `[ة] → ه`), and non-alphabetic punctuation is removed.
2. **Pericope Segmentation & Token-Jaccard Scoring:** Canonical episode spans (P_1 … P_7) are delimited by zero-based character offsets. Local alignment between a witness span S_w and canonical episode seed tokens S_k is measured by tokenized Jaccard similarity: $$J(S_w, S_k) = \frac{|T(S_w) \cap T(S_k)|}{|T(S_w) \cup T(S_k)|}$$ where $T(S)$ represents the unique token set of span S. Compiler abridgment notices are classified as structural references rather than textual omissions, preventing artificial penalties against editorial summaries.

3. **Common Link Confidence Formulation:** In Riwaq's implementation (`clpcl-analyzer.js`), candidate common-link confidence $\text{Conf}(v)$ is calculated via a multi-signal bounded linear function: $\text{Conf}(v) = \text{clamp}(0.30 \cdot S_1(v) + 0.25 \cdot S_2(v) + 0.15 \cdot S_3(v) + 0.20 \cdot S_4(v) + 0.10 \cdot S_5(v) - 0.20 \cdot P_1(v) - 0.10 \cdot P_2(v) - 0.05 \cdot P_3(v), 0, 1)$ where $\operatorname{norm}(x, \text{lo}, \text{hi}) = \operatorname{clamp}\left(\frac{x - \text{lo}}{\text{hi} - \text{lo}}, 0, 1\right)$ and $\text{clamp}(x, 0, 1) = \min(\max(x, 0), 1)$. The constitutive signals ($S_1 \ldots S_5$) and negative penalties ($P_1 \ldots P_3$) represent:
    - $S_1(v)$ = $\text{norm}(\text{fan_out}(v), 3, 8)$: normalized count of immediate pupil lines radiating from node v;
    - $S_2(v)$ = $\text{bundle_coverage}(v)$: proportion of total bundle compiler paths traversing node v;
    - $S_3(v)$ = $\text{norm}(\text{collector_diversity}(v), 2, 8)$: normalized count of distinct canonical compilers receiving reports through v;
    - $S_4(v)$ = $\text{pre_single_strand_ratio}(v)$: proportion of pre-CL authorities exhibiting single-strand transmission (0.0 for al-Shaʿbī);
    - $S_5(v)$ = $\text{matn_coherence}(v)$: mean pairwise token-Jaccard similarity across aligned pericopes among v's pupil branches (falling back to a 0.50 constant when unsegmented);
    - $P_1(v)$ = $\text{bypass_ratio}(v)$: proportion of bundle paths reaching earlier authorities without traversing v;
    - $P_2(v)$ = $\text{chronology_conflict_ratio}(v)$: proportion of transmission links violating biographical death-date sequencing (0.0 in this corpus);
    - $P_3(v)$ = 1 − $\text{provenance_completeness_ratio}(v)$: penalty for unverified biographical metadata (0.0 in this audited corpus).

The step-by-step mathematical decomposition of al-Shaʿbī's confidence score across all three analytical runs is detailed below:

Scoring Component	Mathematical Formulation	Weight / Scale	Baseline Unsegmented [13]	Masar-Enriched Baseline [15]	Expanded 20-Witness [15]
Fan-out (S_1)	$\text{norm}(\text{fan_out}, 3, 8)$	+0.30	1.0000 × 0.30 = 0.3000	1.0000 × 0.30 = 0.3000	1.0000 × 0.30 = 0.3000
Bundle Coverage (S_2)	$\text{paths}(v)/\text{total_paths}$	+0.25	0.8500 × 0.25 = 0.2125	0.8500 × 0.25 = 0.2125	0.8636 × 0.25 = 0.2159
Collector Diversity (S_3)	$\text{norm}(\text{collectors}, 2, 8)$	+0.15	1.0000 × 0.15 = 0.1500	1.0000 × 0.15 = 0.1500	1.0000 × 0.15 = 0.1500
Pre-Single Strand (S_4)	$\text{pre_single_strand_ratio}$	+0.20	0.0000 × 0.20 = 0.0000	0.0000 × 0.20 = 0.0000	0.0000 × 0.20 = 0.0000
Matn Coherence (S_5)	Mean pairwise token-Jaccard	+0.10	0.5000 × 0.10 = 0.0500	0.1720 × 0.10 = 0.0172	0.1747 × 0.10 = 0.0175
Bypass Penalty (P_1)	$\text{bypass_paths}/\text{total_paths}$	−0.20	0.0500 × −0.20 = −0.0100	0.0500 × −0.20 = −0.0100	0.0455 × −0.20 = −0.0091
Chronology Penalty (P_2)	conflict_ratio	−0.10	0.0000 × −0.10 = 0.0000	0.0000 × −0.10 = 0.0000	0.0000 × −0.10 = 0.0000
Provenance Penalty (P_3)	1 − $\text{completeness_ratio}$	−0.05	0.0000 × −0.05 = 0.0000	0.0000 × −0.05 = 0.0000	0.0000 × −0.05 = 0.0000

Scoring Component	Mathematical Formulation	Weight / Scale	Baseline Unsegmented [13]	Masar-Enriched Baseline [15]	Expanded 20-Witness [15]
Total Confidence Conf (*v*)	clamp $(\sum S_i - \sum P_j, 0, 1)$	[0, 1]	**0.7025**	**0.6697**	**0.6743**

Exact unrounded arithmetic verification (Expanded Model): $$\begin{aligned} \operatorname{Conf}(v) &= 0.30 \cdot S_1 + 0.25 \cdot S_2 + 0.15 \cdot S_3 + 0.20 \cdot S_4 + 0.10 \cdot S_5 - 0.20 \cdot P_1 - 0.10 \cdot P_2 - 0.05 \cdot P_3 \\ &= 0.30 \cdot (1.0) + 0.25 \cdot \left(\frac{19}{22}\right) + 0.15 \cdot (1.0) + 0.20 \cdot (0.0) + 0.10 \cdot (0.17465276) \\ &\quad - 0.20 \cdot \left(\frac{1}{22}\right) - 0.10 \cdot (0.0) - 0.05 \cdot (0.0) \\ &= 0.300000 + 0.215909 + 0.150000 + 0.000000 + 0.017465 - 0.009091 - 0.000000 - 0.000000 \\ &= 0.674283 \longrightarrow \mathbf{0.6743} \end{aligned}$$

Reconciliation of computational scores: In earlier draft summaries, an intermediate unsegmented estimate (S_5 = 0.1805) was inadvertently retained in the table text, which yielded 0.6749 upon post-hoc addition (0.3000 + 0.2159 + 0.1500 + 0.0181 − 0.0091 = 0.6749). The canonical execution of Riwaq's `clpcl-analyzer.js` on the frozen 20-witness dataset (`jassasa/jassasah-masar-20witness-import.json`) measures an exact empirical token-Jaccard matn coherence across the 22 paths of S_5 = 0.17465275871 (≈ 0.1747). As shown above, this yields raw Conf (*v*) = 0.674283457689, which rounds to **0.6743**, matching Riwaq's machine receipt exactly with zero discrepancy.

Computational Reproducibility Artifacts and Verification Audit

All underlying computational datasets, graph schemas, and analyzer scripts are permanently archived across the project repositories: 1. **Primary Analytical Dataset:** `jassasa/jassasah-masar-20witness-import.json` (contains the twenty audited records, twenty-two compiler paths, normalized isnād chains, zero-based pericope character offsets for P_1 … P_7, and full primary Arabic transcriptions). This dataset originates in the upstream curation repository **Masar** (github.com/munir-abbasi/masar/blob/main/jassasa/jassasah-masar-20witness-import.json) and is integrated into the downstream analysis repository **Riwaq** (github.com/munir-abbasi/riwaq/blob/main/jassasa/jassasah-masar-20witness-import.json). 2. **Canonical Engine Execution Report:** `jassasa/jassasah-riwaq-export-20witness.md` (github.com/munir-abbasi/riwaq/blob/main/jassasa/jassasah-riwaq-export-20witness.md), storing the unrounded metric receipts, graph topology features, and candidate outcome classifications produced by Riwaq. 3. **Execution Script & Formula Implementation:** The algorithm is implemented in Riwaq's core analysis module (`scripts/clpcl-analyzer.js`). To reproduce the results, researchers may use the official Riwaq CLI pipeline interface or the audited ECMAScript module invocation: ```bash # Method A: Official Riwaq CLI interface (pipeline analysis and report export) node scripts/riwaq-cli.mjs analyze jassasa/jassasah-masar-20witness-import.json node scripts/riwaq-cli.mjs export jassasa/jassasah-masar-20witness-import.json –format md –out jassasa/jassasah-riwaq-export-20witness.md

Method B: Direct programmatic evaluation using Riwaq's exported ESM analyzeBatch API node –input-type=module -e ' import fs from "fs"; import { analyzeBatch } from "./scripts/clpcl-analyzer.js"; const batch = JSON.parse(fs.readFileSync("jassasa/jassasah-masar-20witness-import.json", "utf8")); const res = analyzeBatch(batch); const cl = res.candidates.find(c => c.narrator_id === "amir-al-shabi"); console.log(`al-Shaʿbī Confidence: ${cl.display_confidence_4dp} (Raw: ${cl.final_confidence}) | Matn Coherence (S5): ${cl.features.matn_coherence}`); ' ``` 4. **Source Edition Verification Audit:** The complete witness-level verification audit (jassasa/jassasah-source-audit.md and jassasa/jassasah-edition-manuscript-verification.md`), detailing 100% of the twenty primary records (twenty-two compiler paths) cross-checked against digitized text transcriptions of standard reference printings (Sunnah.com, Islamweb

Library, Shamela, and Wikisource, reflecting standard scholarly editions such as Fuʾād ʿAbd al-Bāqī's *Ṣaḥīḥ Muslim*, Muḥyī al-Dīn ʿAbd al-Ḥamīd's *Sunan Abī Dāwūd*, Aḥmad Shākir's *Jāmiʿ al-Tirmidhī*, Saʿd al-Shathrī's *Muṣannaf Ibn Abī Shaybah*, Shuʿayb al-Arnāʾūṭ's *Ṣaḥīḥ Ibn Ḥibbān*, and Ḥamdī ʿAbd al-Majīd al-Salafī's *al-Muʿjam al-Kabīr*), is permanently deposited alongside the dataset in the public repository archive (github.com/munir-abbasi/masar/blob/main/jassasa/jassasah-source-audit.md). Physical ink-on-paper volumes and manuscript folios were not directly examined in this study.

Epistemic boundary of the metric: The coefficients above ($w_f = 0.30, w_c = 0.25, w_d = 0.15, w_s = 0.20, w_m = 0.10, w_b = 0.20$) represent Riwaq's calibrated engineering defaults for hadith graph exploration. They provide a transparent, reproducible multi-signal heuristic for ranking candidate transmission nodes, rather than an axiomatic probability density function or statistical estimator. Machine-readable datasets, character spans, and network graphs are archived in JSON format [15].

The observed transmission network

The dominant configuration is Fāṭimah bint Qays → ʿĀmir al-Shaʿbī → several reported pupils → later transmitters → compilers. The nine al-Shaʿbī pupil labels in the encoded corpus are ʿAbd Allāh ibn Buraydah, Sayyār Abū al-Ḥakam, Ghaylān ibn Jarīr, Abū al-Zinād, Qatādah, Dāwūd ibn Abī Hind, Mujālid ibn Saʿīd, Muḥammad ibn Ayyūb Abū ʿĀṣim, and al-Shaybānī. The last two labels remain subject to the source-level identity qualifications recorded in the audit.

The Biographical Anchor: ʿĀmir ibn Sharāḥīl al-Shaʿbī (d. ca. 103–105 AH / 721–723 CE)

The structural emergence of ʿĀmir al-Shaʿbī as the principal structural common-link candidate is historically and philologically profound. Al-Shaʿbī was the foremost jurist and traditionalist of Kufa at the turn of the eighth century CE. Born during the caliphate of ʿUmar ibn al-Khaṭṭāb (ca. 20–31 AH), he lived at the intellectual and political crossroads of the early Islamic empire, serving as an advisor and diplomatic envoy for the Umayyad caliph ʿAbd al-Malik ibn Marwān to the Byzantine court in Constantinople, and subsequently as *qāḍī* of Kufa appointed by ʿUmar ibn ʿAbd al-ʿAzīz [12]. Al-Shaʿbī is reported to have encountered upwards of five hundred Companions of the Prophet, narrating directly from several dozen, including early female authorities such as ʿĀʾishah and Fāṭimah bint Qays.

Significantly, al-Shaʿbī was renowned for an uncompromising oral memory ethos: classical biographical compendia famously record his declaration, *"I never wrote black on white [ink on parchment], nor did any man narrate a hadith to me that I ever desired him to repeat"* (*mā katabtu sawādan fī bayāḍ...*) [11, 12]. His transmission circle functioned as an inter-regional transit conduit through which Kufan legal and narrative traditions were disseminated across the early Islamic provinces. The nine pupil lines identified in this dataset —spanning Basran ascetics and traditionists like Ghaylān ibn Jarīr, Qatādah, and Dāwūd ibn Abī Hind, to prominent Kufan specialists like Mujālid ibn Saʿīd and Sayyār Abū al-Ḥakam—demonstrate that his teaching assemblies operated as a widespread supra-regional clearinghouse rather than a localized provincial circle.

This fan-out is visible independently of the Riwaq score. The Ibn Buraydah–Ḥusayn al-Muʿallim–ʿAbd al-Wārith line connects Muslim 2942a with Abū Dāwūd 4326 [1, 2]. Sayyār's line through Qurrah ibn Khālid appears in Muslim 2942b, al-Ṭayālisī, and al-Ṭabarānī 968 [1, 9, 10]. Ghaylān and Abū al-Zinād are represented in Muslim's abbreviated notices [1]. The Qatādah branch appears in al-Tirmidhī [3]; the Dāwūd–Ḥammād line in Aḥmad and al-Nasāʾī [5, 6]; and Mujālid's lines in Abū Dāwūd, Ibn Mājah and Ibn Abī Shaybah [2, 4, 7]. These repeated lower chains are observations of transmitted relationships. In isnād-cum-matn analysis, these pupil lines represent downstream partial common links or single strands radiating from

al-Shaʿbī; while—under the standard working premise of the method—they indicate that the narrative was circulating in al-Shaʿbī's Kufan milieu, they converge on him and cannot be treated as separate, independent lines reaching back to Fāṭimah or the Prophet.

The broader Fāṭimah network also contains Abū Salamah → al-Zuhrī, attested in Abū Dāwūd 4325, and Yaḥyā ibn Yaʿmar → Ibn Buraydah, attested in Ibn Ḥibbān 6787 [2, 8]. Thus, while al-Shaʿbī is central to the major family sampled, he does not exhaust the reported pathways from Fāṭimah. These two non-Shaʿbī routes constitute genuine collateral lines from Medina and Khurasan/Basra, showing that knowledge of Fāṭimah's tradition was not confined to al-Shaʿbī's Kufan circle. Ibn Buraydah's appearance in both the al-Shaʿbī and Yaḥyā routes also complicates an overly simple picture of separate lines. Abū Dāwūd 4328, through Jābir and Abū Salamah, is retained as a thematic comparator rather than merged into Fāṭimah's formal isnād [2].

Within these branches further subdivisions can be observed. Qurrah leads to three sampled pupils; Mujālid to Ismāʿīl ibn Abī Khālid and Abū Usāmah; and the latter Ismāʿīl line to different later transmitters. This structure helps formulate hypotheses about where textual clusters may have been transmitted. It describes transmission branching, not eyewitness verification.

Variation in the narrative frame and the voyagers

The sermon occasion is unstable across the displayed readings. Muslim's expansive version situates the episode within Fāṭimah's personal account and a smiling public address; other notices place the summons after a delayed evening prayer or an unusual ẓuhr ascent to the pulpit. Ibn Mājah emphasizes the Prophet's pleasure at the news and a missed midday rest [1, 2, 4]. Such differences might reflect abbreviated narration, alternative framing or repeated retelling. The corpus cannot settle which explanation is correct.

The identity of those who made the voyage is even more consequential, exhibiting three distinct narrative profiles: 1. **Tamīm sailing personally with companions:** Muslim 2942a and Abū Dāwūd 4326 (via Ibn Buraydah) explicitly record that Tamīm narrated that he embarked in a sea vessel with thirty men of Lakhm and Judhām (*rakiba fī safīnah baḥriyyah maʿa thalāthīna rajulan min Lakhm wa-Judhām*). The Dāwūd ibn Abī Hind branch (al-Nasāʾī 4453, Aḥmad 26066, al-Ṭabarānī 964) similarly records Tamīm sailing with his companions (*aṣḥāb*). In an abbreviated formulation, Ghaylān ibn Jarīr (Muslim 2942c) likewise preserves Tamīm's personal landing on the island seeking water (*fasaqaṭa ilā jazīrah fa-kharaja ilayhā yaltamisu al-māʾ*). 2. **Tamīm's kinsmen sailing without him:** Muslim 2942b (via Sayyār) states that kinsmen of Tamīm took to the sea (*inna banī ʿamm li-Tamīm al-Dārī rakibū fī al-baḥr*), echoed in al-Ṭayālisī 1751 and al-Ṭabarānī 968. Muḥammad ibn Ayyūb Abū ʿĀṣim (al-Ṭabarānī 956) and al-Shaybānī (al-Ṭabarānī 960) likewise preserve the kinsmen framing (*banū ʿamm lah* / *nāsan min banī ʿamm Tamīm*). Within the Mujālid branch, Ibn Mājah 4074 reads that a cousin of Tamīm informed him (*ibn ʿamm li-Tamīm akhbaranī*), while Ibn Abī Shaybah 40425 records a party of his kinsmen (*rahṭan min banī ʿammih*). 3. **Unnamed regional voyagers:** Al-Tirmidhī 2253 (via Qatādah) describes people from the people of Palestine (*nāsan min ahl Filasṭīn*) boarding a ship, without identifying Tamīm as one of the passengers; and Abū Dāwūd 4325 (via Abū Salamah) presents Tamīm recounting news from an unnamed man on an island (*ʿan rajul kāna fī jazīrah*).

In classical *jarḥ wa-taʿdīl* literature, Mujālid ibn Saʿīd was frequently characterized as weak (*ḍaʿīf*) and prone to confused transmission (*taghayyur* / *awhām*) [11], prompting commentators and early takhrīj analyses to suspect that attributing the voyage to a cousin or kinsmen was an idiosyncratic defect introduced by Mujālid. However, the multi-branch evidence demonstrates that kinsmen language (*banū ʿamm*) is attested across four distinct pupil lines of al-Shaʿbī: Sayyār, Muḥammad ibn Ayyūb, al-Shaybānī, and Mujālid. Consequently, kinsmen attribution alone is **not** a diagnostic marker of Mujālid. A narrower grouping of the Mujālid readings

can be made from the *combination* of a kinsmen account with the distressed captive, a question about the Arabs' religious condition, and a particular sequence of the geographical questions [4, 7]. Even that grouping remains a textual hypothesis pending a larger collation.

The Jassāsah and the captive

Collation of the encountered entity across the twenty-witness corpus reveals three distinct lexical registers, rather than a uniform binary:

1. **The Zoomorphic Beast (*dābbah* / *al-Jassās*):** In Muslim 2942a (Ibn Buraydah), the figure is "دابة أهلب كثير الشعر", a heavily hairy creature whose front and back cannot readily be distinguished [1]. This zoomorphic designation (*dābbah*) is preserved in Muslim 2942b (Sayyār), al-Ṭayālisī 1751, al-Ṭabarānī 968, al-Tirmidhī 2253 (Qatādah), and the Dāwūd ibn Abī Hind branch (al-Nasāʾī 4453, Aḥmad 26066, al-Ṭabarānī 964). Al-Ṭabarānī 967 adds the unique masculine nominal form *al-Jassās* [1, 3, 5, 6, 9, 10, 14]. Importantly, *al-Jassās* in al-Ṭabarānī 967 functions as a masculine nominal designation rather than an explicit zoological description; grammatical form (masculine agent noun) and inferred creature morphology must therefore be treated as distinct analytical variables in textual collation.
2. **The Ambiguous Hairy Human or Shape (*insān* / *shayʾ* / *shabaḥ*):** A significant cluster within al-Shaʿbī's Kufan transmission circle describes the entity not as an animal, but as a hair-shrouded human or indefinite entity whose gender is visually obscured. Crucially, Muslim 2942c (Ghaylān ibn Jarīr → al-Shaʿbī) explicitly records that Tamīm landed on an island seeking water and *"met a human dragging his hair"* («فَلَقِيَ إِنْسَانًا يَجُرُّ شَعَرَهُ») [1]. Similarly, al-Ṭabarānī 960 (al-Shaybānī → al-Shaʿbī) records: *"they came upon a human, they did not know whether male or female from the profusion of hair"* («فَإِذَا هُمْ بِإِنْسَانٍ لَا يَدْرُونَ أَذَكَرٌ هُوَ أَمْ أُنْثَى مِنْ كَثْرَةِ الشَّعْرِ») [10]. Al-Ṭabarānī 956 (Abū ʿĀṣim → al-Shaʿbī) likewise reads *"they met an entity, they did not know whether a man or a woman"* («فَلَقُوا شَيْئًا لَا يَدْرُونَ رَجُلًا هُوَ أَوْ امْرَأَةً») [10]. This ambiguous human register is also shared by the Mujālid branch: Ibn Mājah 4074 reads *"a shaggy human form dragging his hair"* («شَبَحًا يَجُرُّ شَعَرَهُ»), while Ibn Abī Shaybah 40425 has *"a black, shaggy thing"* («شَيْئًا أَسْوَدَ أَهْلَبَ») [4, 7].
3. **The Explicit Female Human (*imraʾah* / *jāriyah*):** Outside al-Shaʿbī's circle, the independent collateral lines explicitly introduce female human personifications: Abū Dāwūd 4325 (via Abū Salamah in Medina) reads "بامرأة تجر شعرها", a woman dragging her hair [2], while Ibn Ḥibbān 6787 (via Yaḥyā ibn Yaʿmar in Khurasan/Basra) reads "جارية تجر شعرها", a young woman or maiden dragging her hair [8]. In Abū Dāwūd 4328 (the Jābir comparator), the female characterization is supplied in a reported transmitter's explanatory dialogue rather than the primary encounter narrative [2].

These textual alignments disprove any claim of rigid zoomorphic uniformity in al-Shaʿbī's Kufan circle: al-Shaʿbī's pupils oscillate between the zoomorphic beast (*dābbah*) and the hair-covered human or entity (*insān* / *shayʾ*). Within the present sampled corpus, the *explicitly female personification* (*imraʾah* / *jāriyah*) is uniquely concentrated in the non-Shaʿbī collateral lines of Medina and Khurasan; whether this concentration reflects an early regional differentiation or an artifact of surviving textual selection remains an open question for broader manuscript collation.

Abū Dāwūd 4328 is revealing about attribution levels. A female characterization of al-Jassāsah is supplied in a reported transmitter's explanatory dialogue, rather than appearing as the same sentence of the island encounter as in 4325 [2]. That explanatory gloss is evidence about reception and interpretation, but should not automatically be inserted into the original speech of a narrator in every other version.

The captive likewise ranges from an unusually large, tightly bound man to an elderly, distressed figure or a man whose motion or leap inspires fear. Variation is found both in descriptive vocabulary and in sequence: some readings identify him as the Dajjāl immediately; others place substantial questioning before the identification [1–8]. No single term should be harmonized with another without a separate lexical and manuscript inquiry.

Questions, order, and additions

The fullest Muslim account arranges the inquiries about the date palms of Baysān, Lake Ṭabariyyah, the spring of Zughar, and the Arabian Prophet in that order [1]. But the order is not invariant. The compact al-Tirmidhī recension asks first about Zughar and then a lake and Baysān [3]. Ibn Mājah and Ibn Abī Shaybah place questions about the Prophet and Arabs before Zughar, Baysān and Ṭabariyyah [4, 7]. The Dāwūd branch in Aḥmad and al-Nasāʾī places the prophetic question and a question about Persia before the geographic sequence [5, 6]. The consulted al-Ṭabarānī witnesses add further permutations [10].

These observations show both an identifiable shared story and considerable freedom in how its interrogation scene is organized. The question about Persia is demonstrably branch-concentrated in the Dāwūd–Ḥammād reports (*al-Sunan al-Kubrā* 4453, *Musnad Aḥmad* 26066, *al-Muʿjam al-Kabīr* 964). However, the precise historical moment and mechanism of its emergence remain unknown: it could reflect an interpolation or regional gloss during mid-second-century Basran transmission (e.g., around Ḥammād ibn Salamah, d. 167 AH), or alternatively an authentic early variant not preserved in other branches; the surviving digital evidence establishes its branch concentration but cannot fix its exact date of origin.

Similarly, when the Mujālid cluster (*Sunan Ibn Mājah* 4074, *Musannaf Ibn Abī Shaybah* 40425, *Sunan Abī Dāwūd* 4327) is compared systematically against all other sampled branches, its distinctive profile emerges not from the kinsmen reading alone—which is shared with three other al-Shaʿbī lines (Sayyār, Abū ʿĀṣim, al-Shaybānī)—but from a unique *constellation of co-occurring narrative features*: (1) the kinsmen framing (P_2), (2) the ambiguous human or shaggy form (P_3, shared with Ghaylān, Abū ʿĀṣim, and al-Shaybānī, but absent from Ibn Buraydah, Qatādah, and Dāwūd), (3) an inverted interrogation sequence placing the Prophet and the political-religious description of Arab unity before the geographical inquiries (P_5), and (4) an abbreviated conclusion omitting the extended geographical sanctuary perimeter. This multi-feature alignment demonstrates that Mujālid's branch represents a distinct recensional sub-tradition within Kufa, rather than a single isolated scribal error. Neither pattern proves that these features originated with the named transmitters, but cladistic tracking across the Riwaq graph reveals that interrogation sequence acted as a flexible oral-transmission variable: even within a shared lower line (such as Sayyār–Qurrah), sequence diverges, indicating that order was not frozen at the Common Link stage [13, 15].

The protection of Ṭaybah/Medina and, in some versions, Mecca recurs, but endings differ in their elaboration of guarding angels and the Prophet's gesture indicating an eastern direction [1–4, 10]. The shorter texts must be read with their own abridgment notices in view. Absence from an abridged source is not equivalent to demonstrated omission from the transmitter's entire account.

The related Jābir report

The Jābir witness in Abū Dāwūd 4328 presents sailors who exhaust their provisions, an encounter with al-Jassāsah, and a further dialogue identifying the Dajjāl with Ibn Ṣayyād [2]. This is a genuine companion-level difference within a connected narrative field. The Ibn Ṣayyād identification is reported as an attributed claim

within the Jābir transmission, not as a historical conclusion endorsed by the present investigation. Counting this notice as another independent copy of the long Fāṭimah sermon would erase its different narrative frame and appendices. Keeping it as a comparator preserves both similarity and divergence.

What Riwaq establishes, and what its scores cannot establish

Methodological Triad: In evaluating hadith transmission networks, three distinct analytical operations must be rigorously kept separate: 1. **Computational Collation:** Algorithmic alignment, graph generation, and metric calculation (such as pericope span tracking and token-Jaccard coherence), which verify formal textual overlaps and topological densities across indexed primary texts. Automated pericope alignment serves strictly within this category as a scalable computational collation tool that standardizes segment boundaries across voluminous texts, assisting rather than replacing philological analysis. 2. **Philological Interpretation:** Scholarly analysis of lexical registers, syntactic shifts, narrative motifs, and contextual nuances across surviving recensions. 3. **Historical Reconstruction:** Inductive inference regarding transmission horizons, historical provenance, and social context, which remains conditional on the degree to which surviving chains reflect actual transmission history.

The audited import contains eighteen encoded witnesses, twenty compiler-path variants, and seventy-one narrator profiles, with eighty-two graph edges reported in the retained pipeline review. Al-Ṭabarānī 967 initially remained outside the graph because its conjunction had not been resolved. In the initial unsegmented baseline run, Riwaq ranked al-Shaʿbī first with a structural score of **0.7025** (candidate outcome `contested`, fan-out **9**, bundle coverage **85.0%**, collector diversity **9**, bypass ratio **5.0%**), while Fāṭimah ranked second (score **0.4375**, outcome `uncertain`) and Ibn Buraydah third as a partial-common-link candidate (score **0.0900**, outcome `likely_weak_in_context`) [13].

Crucially, in that initial unsegmented run, **no imported variant contained machine-encoded** `matn_segments`, causing Riwaq's `computeNodeMatnCoherence()` to fall back to `default_constant_unmeasured` (a constant 0.50 weight). To overcome this foundational limitation, the Masar Takhrīj Engine's automated pericope alignment engine (`PericopeAligner`) was deployed across seven canonical narrative episodes (P_1 … P_7), injecting verified zero-based character spans directly into the analytical dataset. Re-executing the Riwaq pipeline with active matn segments activated empirical token-Jaccard coherence scoring (`measured_token_jaccard`). Furthermore, promoting two additional discovered witnesses from Masar's verified proposal buffer (al-Ṭabarānī 964 and 967) expanded the model to a 20-witness, 22-path corpus [15].

The comparison across these three analytical configurations demonstrates the empirical effect of machine-measured matn coherence and corpus expansion:

Analytical Configuration	Compiler Paths	Narrators	Matn Coherence Basis	Al-Shaʿbī Confidence	Bundle Coverage	Bypass Ratio	Candidate Outcome
1. Initial Unsegmented Baseline [13]	20	71	`default_constant_unmeasured`	0.7025	85.0%	5.0%	First, `contested`
2. Masar-Enriched 18-Witness [15]	20	71	`measured_token_jaccard`	**0.6697**	85.0%	5.0%	First, `contested`
3. Expanded 20-Witness [15]	22	77	`measured_token_jaccard`	**0.6743**	86.4%	4.5%	First, `contested`

Across both empirical runs (the enriched 18-witness baseline and the expanded 20-witness model), ʿĀmir al-Shaʿbī remains the leading structural common-link candidate. Integrating empirical matn-coherence slightly moderates the confidence score from the artificial 0.7025 constant to an empirically grounded **0.6697** (bundle coverage 85.0%, bypass ratio 5.0%), reflecting observed variance in narrative phrasing across branches. Expanding the corpus to twenty records incorporates the Dāwūd–Ḥammād route in al-Ṭabarānī 964 (with its unique Persia interrogation) and resolves the Qatādah/Ibrāhīm conjunction in al-Ṭabarānī 967 (with the masculine *al-Jassās*), increasing al-Shaʿbī's confidence to **0.6743**, broadening bundle coverage to **86.4%**, and shrinking the bypass ratio to **4.55%** [15]. Fāṭimah bint Qays likewise shifts from an unmeasured 0.4375 to an empirical **0.4048** (18-witness baseline) and **0.4062** (20-record expanded model), remaining in the `uncertain` tier.

Beyond aggregate scores, Masar's episode-level pericope alignment provides a computational test of specific textual attributions. In classical *jarḥ wa-taʿdīl* criticism and early hadith commentary, Mujālid ibn Saʿīd was singled out for erratic memory and unstable phrasings, leading commentators to question whether attributing the voyage to Tamīm's cousin or kinsmen rather than to Tamīm himself was an idiosyncratic corruption originating exclusively with Mujālid [11]. The computational pericope matrix clarifies this transmission problem by revealing that narrative variations in traveler identity (P_2: `TRAVELER_IDENTITY`) are distributed across distinct, recognizable clusters within al-Shaʿbī's transmission circle, as detailed in Table 1:

Table 1: Cross-Branch Attestation of the Kinsmen Voyage (P_2) and Traveler Variations Across al-Shaʿbī's Circle

Category & Transmission Branch	Primary Attestation	Recorded Travel Party Phrasing	Geographic Center
A. Kinsmen / Cousin Cluster (*banū ʿamm* / *ibn ʿamm*)			
1. Sayyār Abū al-Ḥakam	*Musnad al-Ṭayālisī* 1751, *al-Muʿjam al-Kabīr* 968, *Ṣaḥīḥ Muslim* 2942b	"إن بني عم لتميم الداري ركبوا في البحر" (Kinsmen of Tamīm al-Dārī took to the sea)	Kufa / Wasit
2. Muḥammad ibn Ayyūb Abū ʿĀṣim	*al-Muʿjam al-Kabīr* 956	"أن بني عم له ركبوا البحر" (That kinsmen of his took to the sea)	Kufa / Basra
3. al-Shaybānī	*al-Muʿjam al-Kabīr* 960	"أن ناسا من بني عم تميم ركبوا في البحر" (People of Tamīm's kinsmen took to the sea)	Kufa
4. Mujālid ibn Saʿīd	*Sunan Ibn Mājah* 4074, *Musannaf Ibn Abī Shaybah* 40425	"ابن عم لتميم" (A cousin of Tamīm) / "رهطا من بني عمه" (A party of his kinsmen)	Kufa
B. Personal / Regional Voyage Formulations			
5. ʿAbd Allāh ibn Buraydah	*Ṣaḥīḥ Muslim* 2942a, *Sunan Abī Dāwūd* 4326	"ركب في سفينة بحرية مع ثلاثين رجلا من لخم وجذام" (Tamīm sailed with thirty men of Lakhm and Judhām)	Basra / Merv

Category & Transmission Branch	Primary Attestation	Recorded Travel Party Phrasing	Geographic Center
6. Qatādah ibn Diʿāmah	*Jāmiʿ al-Tirmidhī* 2253	"ناسا من أهل فلسطين ركبوا سفينة" (People of Palestine took a ship)	Basra
7. Dāwūd ibn Abī Hind	*al-Sunan al-Kubrā* 4453, *Musnad Aḥmad* 26066, *al-Muʿjam al-Kabīr* 964	Preserves Tamīm embarking with his companions (*aṣḥāb*)	Basra
8. Ghaylān ibn Jarīr	*Ṣaḥīḥ Muslim* 2942c	"فسقط إلى جزيرة فخرج إليها يلتمس الماء" (Tamīm landed on an island seeking water)	Basra

Because the kinsmen reading (*banū ʿamm* / *ibn ʿamm*) is attested across four distinct pupil lines radiating from al-Shaʿbī (Sayyār, Muḥammad ibn Ayyūb, al-Shaybānī, and Mujālid), it was already circulating at the level of al-Shaʿbī (d. ca. 103–105 AH) and cannot be characterized as an isolated invention of Mujālid. Simultaneously, the preservation of the Lakhmid personal voyage through Ibn Buraydah and Dāwūd, alongside Palestinian voyagers in Qatādah, demonstrates that multiple narrative formulations coexisted in early transmission rather than a single monolithic archetype.

The enriched Riwaq reports retain candidate graph metrics, uncertainty exclusions, and `outcome_claim` categories. Each candidate's `NO_EVIDENCE_STRUCTURAL` warning explains the absence of external narrator reliability data. While a permitted export confirms that the structural and matn-segment reports pass the software's deterministic preflight gates, computation alone cannot validate historical reality. Traditional biographical evaluation remains distinct: the consulted *Tahdhīb al-Kamāl* records conflicting verdicts on Mujālid [11], while al-Tirmidhī designates his own transmission *ṣaḥīḥ gharīb* [3]. Neither traditional verdict is replaced by graph algorithms.

From graph description to a discriminating isnād-cum-matn test

The deployment of Masar's Layer 2/3 `PericopeAligner` has realized the transition from unweighted isnād graph description to a systematic episode-level alignment. The narrative was segmented into seven canonical pericopes: P_1 (sermon occasion and assembly), P_2 (traveler identity and voyage), P_3 (island landing and encounter with al-Jassāsah), P_4 (description of the chained captive), P_5 (geographic and religious interrogation), P_6 (Messianic self-identification and sanctuary boundaries), and P_7 (Prophetic ratification, orientation gesture, and closing remarks).

Table 2: Pericope Alignment and Witness Attestation Matrix (P_1 … P_7)

The table below delineates the distribution and status of the seven canonical narrative episodes across representative attestations in the twenty-witness corpus:

Witness Locator	Isnād Branch / Authority	P_1: Sermon Frame	P_2: Traveler Identity	P_3: al-Jassāsah Form	P_4: Chained Captive	P_5: Interrogation Order	P_6: Identification	P_7: Prophetic Ratification
Muslim 2942a	Ibn Buraydah → al-Shaʿbī	Full (Pulpit/ Smile)	30 men (Lakhm/ Judhām)	Beast (*dābbah ahlāb*)	Massive, bound	B → Ṭ → Z → P	Dajjāl / 40 days	Staff tap / East pointer
Muslim 2942b	Sayyār → al-Shaʿbī	Hospitality frame	Kinsmen (*banū ʿamm*)	Beast (*dābbah*)	Distressed figure	B → Ṭ → Z → P	Dajjāl / Sanctuaries	East pointer

Witness Locator	Isnād Branch / Authority	P_1: Sermon Frame	P_2: Traveler Identity	P_3: al-Jassāsah Form	P_4: Chained Captive	P_5: Interrogation Order	P_6: Identification	P_7: Prophetic Ratification
Muslim 2942c	Ghaylān → al-Shaʿbī	Abridged notice	Tamīm personal voyage	Human (*insān* dragging hair)	— (cross-ref)	— (cross-ref)	— (cross-ref)	Abridged ending
Muslim 2942d	Abū al-Zinād → al-Shaʿbī	Abridged notice	Cross-reference	—	—	—	—	Abridged ending
Abū Dāwūd 4325	Abū Salamah (Medina)	Night prayer delay	News of a man	Woman (*imraʾah*)	Bound captive	Condensed sequence	Dajjāl / Sanctuaries	Staff tap / East pointer
Abū Dāwūd 4326	Ibn Buraydah → al-Shaʿbī	Assembly call	30 men (Lakhm/ Judhām)	Beast (*dābbah*)	Chained captive	Standard sequence	Dajjāl / Sanctuaries	East pointer
Abū Dāwūd 4327	Mujālid → al-Shaʿbī	Midday prayer	Cross-reference	—	—	—	—	Abridged notice
Abū Dāwūd 4328	Jābir comparator	Distinct assembly	Sailors / Famine	Glossed as woman	Chained man	Abbreviated inquiry	Ibn Ṣayyād dialogue	Distinct Medinan frame
al-Tirmidhī 2253	Qatādah → al-Shaʿbī	Pulpit / Laughter	Palestine voyagers	Beast (*dābbah*)	Chained captive	Z → Ṭ → B → P	Dajjāl / Sanctuaries	Staff tap / East pointer
Ibn Mājah 4074	Mujālid → al-Shaʿbī	Midday prayer	Kinsman (*ibn ʿamm*)	Shaggy human form	Distressed (*ahlakahu al-ghamm*)	P → Arabs → Z → B → Ṭ	Dajjāl / Sanctuaries	Staff tap / East pointer
al-Nasāʾī 4453	Dāwūd → al-Shaʿbī	Hastened assembly	Tamīm & companions	Beast (*dābbah*)	Chained captive	P → **Persia** → B → Ṭ → Z	Dajjāl / Sanctuaries	Minbar strike / East
Aḥmad 26066	Dāwūd → al-Shaʿbī	Hastened assembly	Tamīm & companions	Beast (*dābbah*)	Chained captive	P → **Persia** → B → Ṭ → Z	Dajjāl / Sanctuaries	Minbar strike / East
Ibn Abī Shaybah 40425	Mujālid → al-Shaʿbī	Midday prayer	Kinsmen voyage	Shaggy figure	Distressed captive	P → Arabs → Z → B → Ṭ	Dajjāl / Sanctuaries	Minbar tap / East pointer
Ibn Ḥibbān 6787	Yaḥyā (Khurasan/ Basra)	Direct questioning	Voyagers	Girl (*jāriyah*)	Chained captive	Two signs inquiry	Release signs	Sanctuary protection
al-Ṭayālisī 1751	Sayyār → al-Shaʿbī	Hospitality frame	Kinsmen (*banū ʿamm*)	Beast (*dābbah*)	Distressed figure	Z → Ṭ → B → P	Dajjāl / Sanctuaries	Minbar tap / East
al-Ṭabarānī 956	Abū ʿĀṣim → al-Shaʿbī	Inverted questioning	Kinsmen (*banū ʿamm lah*)	Ambiguous thing (*shayʾ*)	Bound captive	Standard sequence	Dajjāl / Sanctuaries	Minbar strike / East
al-Ṭabarānī 960	al-Shaybānī → al-Shaʿbī	Serene Medina frame	Kinsmen (*nāsan min banī ʿamm*)	Human (*insān*, ambiguous sex)	Bound captive	Standard sequence	Dajjāl / Sanctuaries	Minbar strike / East

Witness Locator	Isnād Branch / Authority	P_1: Sermon Frame	P_2: Traveler Identity	P_3: al-Jassāsah Form	P_4: Chained Captive	P_5: Interrogation Order	P_6: Identification	P_7: Prophetic Ratification
al-Ṭabarānī 964	Dāwūd → al-Shaʿbī	Assembly call	Tamīm & companions	Beast (*dābbah*)	Chained captive	P → **Persia** → B → Ṭ → Z	Dajjāl / Sanctuaries	Minbar strike / East
al-Ṭabarānī 968	Sayyār → al-Shaʿbī	Hospitality frame	Kinsmen (*banū ʿamm*)	Beast (*dābbah*)	Distressed figure	Standard sequence	Dajjāl / Sanctuaries	Minbar tap / East
al-Ṭabarānī 967	Qatādah / Ibrāhīm	Assembly frame	Voyagers	Masculine (*al-Jassās*)	Bound captive	Follower status inquiry	Dajjāl / Sanctuaries	Minbar strike / East

Key to interrogation sequence (P_5): **B** = Palm trees of Baysān; **Ṭ** = Lake Ṭabariyyah; **Z** = Spring of Zughar; **P** = The Unlettered Prophet of the Arabs; **Persia** = Inquest regarding conquest of Persia.

This automated segmentation enables combinations of textual features to be tested against isnād transmission branches rather than isolated motifs. The kinsmen attribution (*banū ʿamm*) is demonstrated to be a shared al-Shaʿbī circular feature across four distinct pupil lines rather than an idiosyncratic marker of Mujālid. Conversely, a genuine Mujālid cluster can be isolated through the joint co-occurrence of the distressed captive (*rajul kabīr ahlakuhu al-ghamm*), the political-religious Arab unity exchange, repeated sighs, and an inverted interrogation sequence (Prophet and Arabs placed before the geographical sites). Similarly, the Dāwūd–Ḥammād branch (Aḥmad 26066, al-Nasāʾī 4453, and al-Ṭabarānī 964) is computationally corroborated by the exclusive pairing of the interrogation regarding Persia (هل ظهر على فارس؟) with the early positioning of the prophetic question before the geographical sequence. Disagreement within the Sayyār–Qurrah line over question order demonstrates that a common lower route does not enforce invariant sequence [15].

These empirical alignments support a nuanced historical model: (i) an early core narrative circulating at the level of al-Shaʿbī with substantial flexibility in scene ordering and descriptive phrasing; (ii) branch-specific stabilizers such as the Dāwūd–Ḥammād Persian addition and the Mujālid descriptive glosses; and (iii) selective compiler abridgment rather than progressive linear accretion.

Discussion

The sampled witnesses establish a recognizable island-and-Dajjāl narrative across multiple transmission configurations. Al-Shaʿbī is the central branching node of the dominant Fāṭimah family under both structural graph analysis and empirical matn-coherence testing; the independently named Abū Salamah and Yaḥyā ibn Yaʿmar approaches prevent an exclusive account of its transmission. The Jābir narrative deserves comparison at the motif level without collapsing its distinct companion attribution and Ibn Ṣayyād dialogue into Fāṭimah's sermon. Repetition across compilers frequently reflects shared lower chains rather than multiple independent early controls [1–10].

The integration of Masar's automated pericope alignment with Riwaq's `measured_token_jaccard` scoring provides a practical computational enhancement over initial unmeasured runs, standardizing episode segmentation across the corpus. With al-Shaʿbī achieving an empirical Common Link confidence of 0.6697 (19-attestation baseline) and 0.6743 (20-record expanded corpus), the computational data substantiate that a cohesive narrative tradition was transmitted through his circle. Furthermore, demonstrating that the

kinsmen voyage spans four distinct pupil lines (Sayyār, Muḥammad ibn Ayyūb, al-Shaybānī, and Mujālid) demonstrates the utility of multi-branch pericope alignment to adjudicate transmission controversies on an empirical basis.

The Historical Meaning of the Common Link: Transmission Horizons, Terminus Distinctions, and the Single Strand

In the methodological framework of Isnād-cum-Matn analysis (ICMA) pioneered by Joseph Schacht [16] and substantially refined by G.H.A. Juynboll [17], Harald Motzki [18], and Gregor Schoeler [19], the identification of a structural Common Link carries specific chronological and historiographical significance:

1. **Transmission Horizons and Terminus Distinctions:** In strict historical method, four chronological and material levels must be carefully distinguished:
    1. *Physical Manuscript Horizon (Codicological Terminus Ante Quem):* Extant physical manuscript codices and material parchment/paper folios (which for the classical Sunni hadith compendia typically date from the fifth to seventh centuries AH / eleventh to thirteenth centuries CE, alongside earlier fragmentary papyri) establish the material *terminus ante quem*—the point before which surviving copies are physically preserved on material carriers.
    2. *Literary Redaction Horizon (Compilation Terminus Ante Quem):* The third-century redaction and recorded death dates of the compilers (e.g., Muslim ibn al-Ḥajjāj, d. 261 AH / 875 CE; Abū Dāwūd al-Sijistānī, d. 275 AH / 889 CE) mark the scholastic horizon by which the individual traditions were compiled, organized, and disseminated in early scholarly literature.
    3. *Documented Biographical Horizon:* The recorded lifespans of named authorities appearing in the isnād chains (such as ʿĀmir al-Shaʿbī, d. ca. 103–105 AH / 721–723 CE) mark when these figures lived according to classical biographical compendia (*kutub al-rijāl*).
    4. *Reconstructed Transmission Horizon (Analytical Terminus a Quo for Surviving Branches):* Within the internal logic of Isnād-cum-Matn analysis, the structural convergence of nine distinct transmission lines upon al-Shaʿbī establishes an analytical transmission horizon. This indicates that—**if the attributed transmission relationships substantially reflect actual historical transmission**—the core narrative (encompassing the maritime voyage, the encounter with al-Jassāsah, the chained captive on the island, the fivefold geographical inquiries, and the sanctuary protection of Mecca and Medina) was in active public circulation in Kufa by the late first century AH (c. 75–100 AH / 695–718 CE). While this finding presents an empirical challenge to late Abbasid dating within the multi-branch transmission record, it remains methodologically conditional upon the premise that isnāds were not systematically back-projected downstream onto al-Shaʿbī.
2. **Common Link vs. Invention (The Historiographical Boundary):** Crucially, within contemporary ICMA methodology, identifying a Common Link does **not** equate to proving that this individual invented or fabricated the report. While early formulations by Schacht [16] and Juynboll [17] tended to equate the Common Link with the historical creator of a tradition, Harald Motzki's corrective demonstrated that a Common Link simply marks the earliest traditionist whose systematic teaching and dissemination resulted in surviving multi-branch transmission lines [18]. Whether al-Shaʿbī received the report orally from Fāṭimah bint Qays, an unnamed intermediary, or regional storytelling circles remains an unproven attribution claim. Graph topology alone cannot verify whether Tamīm al-Dārī experienced an actual maritime event or encountered a mythical beast; such historical questions lie beyond the evidentiary ceiling of isnād-cum-matn analysis.

3. **The Single Strand (SS) and Collateral Transmission:** In ICMA theory, the single strand connecting the Common Link to earlier authorities (Fāṭimah bint Qays → al-Shaʿbī) constitutes an *attribution claim* preserved by the Common Link. Because all downstream transmission lines in this cluster collapse into the bottleneck of al-Shaʿbī, downstream pupil branches do not provide independent eyewitness verification back to the Prophet. However, the survival of collateral, non-Shaʿbī lines through Abū Salamah in Medina (*Abū Dāwūd 4325*) and Yaḥyā ibn Yaʿmar in Khurasan (*Ibn Ḥibbān 6787*) supports an alternative multi-regional transmission history rather than an exclusive Kufan invention by al-Shaʿbī, though these collateral paths are fewer in number and remain dependent on the integrity of their own transmission records.
4. **Oral Fluidity vs. Monolithic Archetype:** The cross-branch distribution of traveler profiles (P_2) fundamentally clarifies the tradition's early evolution. Because four distinct pupil lines transmit the kinsmen reading (*banū ʿamm*), while Ibn Buraydah and Dāwūd preserve the Lakhmid personal voyage, the reconstructive model of ICMA suggests that both variants were circulating within al-Shaʿbī's broader teaching circle in Kufa by the late first century AH, refuting the suspicion that the kinsmen framing was an idiosyncratic corruption of Mujālid. Whether al-Shaʿbī personally varied his wording across different teaching sessions or whether his pupils introduced minor phrasing adaptations during oral transmission remains a conditional historical inference. Nonetheless, the evidence establishes that early transmission possessed natural oral flexibility—such as varying question orders, traveler descriptions, and lexical shifts—characteristic of early Islamic memory culture [19], rather than existing as a rigid, monolithic written archetype.

Future Research Horizons Enabled by Pericope Alignment

The machine-segmented pericope matrix constructed in this study unlocks three immediate lines of inquiry for subsequent computational investigation. First, analyzing the directional polarization in P_7 (the Syrian Sea vs. Yemeni Sea binary vs. the Eastern pointer) can clarify how early Umayyad naval geopolitics intersected with Khurasani eschatological paradigms. Second, isolating the Persia question in P_5 across the Dāwūd–Ḥammād sub-tree provides a potential chronological benchmark for assessing mid-second-century Basran redactions during the early Abbasid period. Finally, computing formal compiler abridgment ratios ($A = \sum \text{len}(P_i) / \text{len}(\text{Pericope Universe})$) across collectors like Muslim and Abū Dāwūd provides a descriptive index of relative textual completeness across recensions rather than an infallible determinant of authorial intention, replacing subjective arguments from silence with measurable editorial entropy benchmarks.

Conclusion

What the sources directly establish. Within nineteen retained digital source attestations (expanded to twenty records yielding twenty-two compiler paths in the supplementary discovery), versions of the Jassāsah story share an identifiable maritime encounter while differing in sermon frame, traveler attribution, description of the encountered being, interrogation content and order, and explanatory additions. The dominant Fāṭimah–al-Shaʿbī transmission fans out through nine labeled pupil branches, alongside two other reported approaches to Fāṭimah and a distinct Jābir comparator [1–10, 15].

What the isnād-cum-matn pattern suggests. Al-Shaʿbī is the leading structural Common Link candidate across all tested models, with an empirical Isnād-cum-Matn confidence of **0.6697** in the 19-attestation baseline and **0.6743** in the expanded 20-record corpus under Riwaq's `measured_token_jaccard` profile. Narrative variation in traveler identity reveals multiple early formulations: the kinsmen voyage (*banū ʿamm*) is computationally confirmed across four distinct pupil branches radiating from al-Shaʿbī, refuting its exclusive attribution to Mujālid, while the Lakhmid companion voyage coexists on parallel lines (Ibn

Buraydah, Dāwūd). Distinctive additions—such as the Persia interrogation in the Dāwūd–Ḥammād line—demonstrate branch-concentrated textual distributions across the surviving corpus, though determining their exact historical moment of authorial inception remains an analytical inference [15].

What could materially change the conclusion. Collation of physical manuscripts and unconsulted early corpora (e.g., al-Ḥumaydī, Isḥāq ibn Rāhwayh, Abū ʿAwānah, al-Bayhaqī, and al-Ṭaḥāwī) could verify whether the Persia question or specific descriptive terms appear in collateral early lines. Independent external dating evidence for named transmitters would refine chronological constraints.

What remains uncertain or unverified. No secure archetypal original text, absolute origination date, or independent historical verification of the island encounter can be deduced from graph topology or pericope alignment. The computational scores describe transmission coherence and branch structure, not historical veracity or report fabrication.

Corpus coverage and limits. Counts apply strictly to the audited digital corpus and its staged expansions in Masar and Riwaq. Full manuscript collation across diverse geographic recensions remains outstanding.

AI Use Disclosure

Artificial Intelligence was utilized in the research, verification of web-based editions, and computational pipeline orchestration for this study. Specifically, a structured prompt was drafted to analyze the initial dataset generated by Riwaq using the Masar Takhrīj Engine. The segmented and enriched dataset produced by Masar was then re-analyzed via Riwaq using Google Gemini Pro, with the resulting metrics, pericope matrices, and algorithmic alignments integrated into the final manuscript to prepare the revised working manuscript.

Note of Caution and Methodological Disclaimer

While this study presents an empirical Isnād-cum-Matn analysis of the *al-Jassāsah* tradition where computational confidence mathematically accounts for exact textual divergence between unabridged reports (*Ṣaḥīḥ Muslim* 2942a, *Sunan Ibn Mājah* 4074) and abridged compiler cross-references (*Ṣaḥīḥ Muslim* 2942b–d, *Sunan Abī Dāwūd* 4327) using Masar and Riwaq, readers are advised to interpret these findings with methodological caution.

Computational isnād graphs and algorithmic pericope alignments produce mathematical models of textual and transmission coherence; they do not generate theological verdicts, nor do they replace traditional Islamic biographical criticism (*ʿilm al-rijāl wa-al-jarḥ wa-al-taʿdīl*). The confidence metrics reflect structural density and lexical overlap within a defined digital sample rather than ontological certainty or historical truth. Consequently, these findings should be critically evaluated, verified, and contextualized by qualified Ḥadīth scholars and classical philologists before being relied upon for normative historical, theological, or legal conclusions.

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